



Simulation of Disc Brake- Thermal

Date: 28 October 2025
Designer: Solidworks
Study name: Transient Thermal
Analysis type: Thermal(Transient)

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Description

Transient Thermal Analysis upto 3 sec. Math Calculated in MATLAB.

Model Information



Model name: kh
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line3 	Solid Body	Mass:10.646 kg Volume:0.00147861 m ³ Density:7,200 kg/m ³ Weight:104.33 N	C:\Users\prati\Downloads\discbrake\brake.SLDPRT May 27 11:10:49 2017
Cut-Extrude2 	Solid Body	Mass:6.89699 kg Volume:0.000957915 m ³ Density:7,200 kg/m ³ Weight:67.5905 N	C:\Users\prati\Downloads\discbrake\njjn.SLDPRT May 27 11:10:49 2017



Study Properties

Study name	Thermal 2
Analysis type	Thermal(Transient)
Mesh type	Solid Mesh
Solver type	FFEPlus
Solution type	Transient
Total time	3 Seconds
Time increment	0.1 Seconds
Contact resistance defined?	No
Result folder	SOLIDWORKS document (C:\Users\prati\Downloads\discbrake)

Units

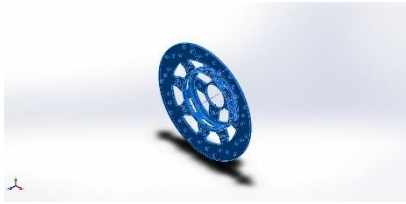
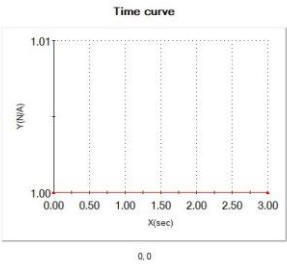
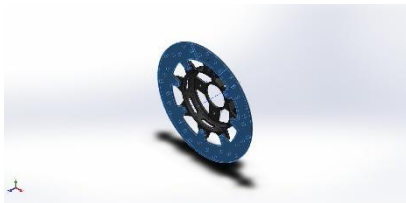
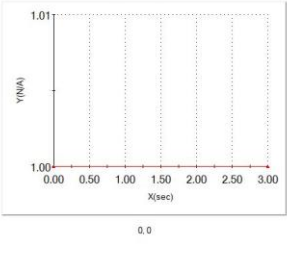
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

Material Properties

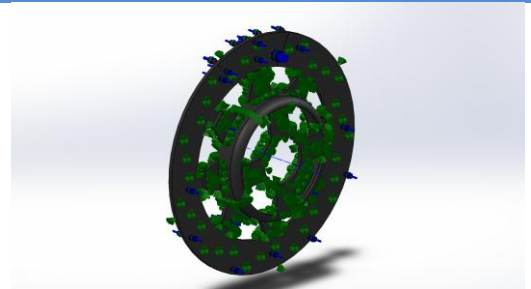
Model Reference	Properties	Components
	Name: Gray Cast Iron Model type: Linear Elastic Isotropic Default failure criterion: Mohr-Coulomb Stress Thermal conductivity: 45 W/(m.K) Specific heat: 510 J/(kg.K) Mass density: 7,200 kg/m ³	SolidBody 1(Split Line3)(brake-1), SolidBody 1(Cut-Extrude2)(njjn-1)
Curve Data:N/A		



Thermal Loads

Load name	Load Image	Load Details	Function Curve
Convection-1		Entities: 196 face(s) Convection 90 Coefficient: $W/(m^2.K)$ Time On variation: Temperature Off variation: Bulk Ambient 293 Kelvin Temperature: Time Off variation:	 Time curve
Heat Power-1		Entities: 4 face(s) Heat Power 6660 W Value: Time on variation:	 Time curve
Temperature-1		Entities: 4 face(s) Initial 293 Kelvin temperature:	

Interaction Information

Interaction	Interaction Image	Interaction Properties
Global Contact		Type: Bonded Components: 1 component(s) Options: Continuous mesh



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	4 Points
Element Size	15.4313 mm
Tolerance	0.771563 mm
Mesh Quality	High
Remesh failed parts independently	Off
Reuse mesh for identical parts in an assembly (Blended curvature-based mesher only)	Off

Mesh information - Details

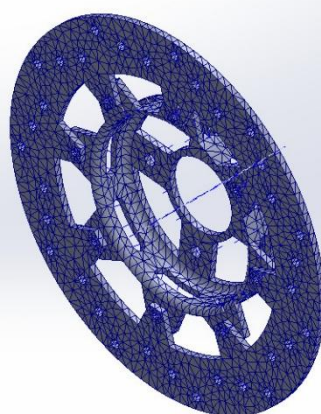
Total Nodes	25353
Total Elements	12262
Maximum Aspect Ratio	13.133
% of elements with Aspect Ratio < 3	85
Percentage of elements with Aspect Ratio > 10	0.0326
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:03
Computer name:	MSI

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Model name: kh
Study name: Thermal 2(-Default-)
Plot type: Mesh Quality1



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Disc Brake- Meshing Quality



Study Results

